

Lesson 3 — Decomposition & Flowcharts

Matches LEGO Teacher Portal lesson: "Forest Flight Rescue".

Objective (I can...): I can break a complex task into sub-problems and represent the plan as a flowchart before I code. **CS concept:** Decomposition and algorithm representation (flowcharts) · **Builds on:** Lesson 2 (sequencing).

AI Connection (2 min): A vision model decomposes "is this a cat?" into layered sub-checks — edges, then shapes, then parts, then the whole. *First-principles question:* "What's the advantage of solving many small sub-problems instead of one big one?"

Materials: LEGO CS & AI kit + device per pair, flowchart symbol reference card (start/end, process, decision, arrow), notebooks. **Model reuse:** Basics model from Lesson 2 — code plus one small build tweak.

60-minute flow - Warm-up (8): "Decompose it" — pairs take a big task ("run a lemonade stand") and break it into three levels of sub-tasks on a whiteboard. Point: decomposition is how engineers make big problems tractable. - **Concept + demo (10):** Introduce flowchart symbols. Coach picks a behavior (patrol forward, then signal), decomposes it aloud, and draws a flowchart — then maps each box to a block. (*Rotate format: "trace it on paper first" — read a flowchart and predict the behavior.*) - **Build/Code (30):** Pairs choose a two-part behavior, decompose it into sub-tasks, draw a flowchart in the notebook, then code it on the built model so blocks match the flowchart. **Make-it-yours beat:** add one attachment and name the build. *Swap at 15 min.* - **Challenge (7):** Add a third sub-task to the flowchart and the code, keeping chart and code in sync. - **Share + pack-up (5):** Pairs show flowchart beside the running program and point out the one-to-one match. Model stays built.

Pair roles: Navigator owns the flowchart; Driver codes from it. *Swap at 15 min. (Trio: third verifies each flowchart box has a matching block.)*

Differentiation: *Support* — a half-drawn flowchart to finish. *Extension / fast-finisher* — decompose a three-part behavior with a sub-task that repeats. *Reteach* — draw one box, code it, confirm, then move to the next box.

Exit check: Pair's flowchart and working program correspond box-for-block. **At-home (optional):** Draw a flowchart for your morning routine with at least six process boxes. ~10 min.